

CLINISCAN: LUNG-ABNORMALITY DETECTION ON CHEST X-RAYS USING AI

-INFOSYS SPRINGBOARD

Milestone 2: Data Preparation & Environment Setup

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1.INTRODUCTION

The objective of Milestone 2 is to develop and implement the core AI components for detecting and classifying lung abnormalities from chest X-ray images. This phase transitions from data preparation to active model development, establishing the baseline performance for the CliniScan system.

In this milestone, we implemented two primary models:

- Detection Model: Designed to identify and locate specific abnormalities using bounding boxes.
- Classification Model: Designed to predict the overarching disease category from the X-ray images

2. DATA DESCRIPTION

The system utilizes the VinDr-CXR dataset, which consists of 15,000 annotated chest X ray images for training.

Key Data Characteristics:

Annotations: Provided in train.csv, containing disease labels and bounding box coordinates. Metadata: Image dimensions (height and width) are stored in

train_meta.csv. Image Format: Images were processed in resized PNG format at a resolution of 1024×1024.

3. Data Preprocessing

To prepare the data for training, several cleaning and transformation steps were performed: Filtering: Removed "No Finding" entries and rows with missing (NaN) bounding boxes to focus the detection model on valid abnormalities.

YOLO Conversion: Bounding boxes were normalized into YOLO format using the equations Empty Labels: Generated empty label files for images without abnormalities to maintain dataset integrity.

Verified dataset integrity

Total images:15000

Total labels:15000

4. Detection Model (YOLOv11)

We utilized the YOLOv11 architecture from Ultralytics for the detection task.

Training Configuration:

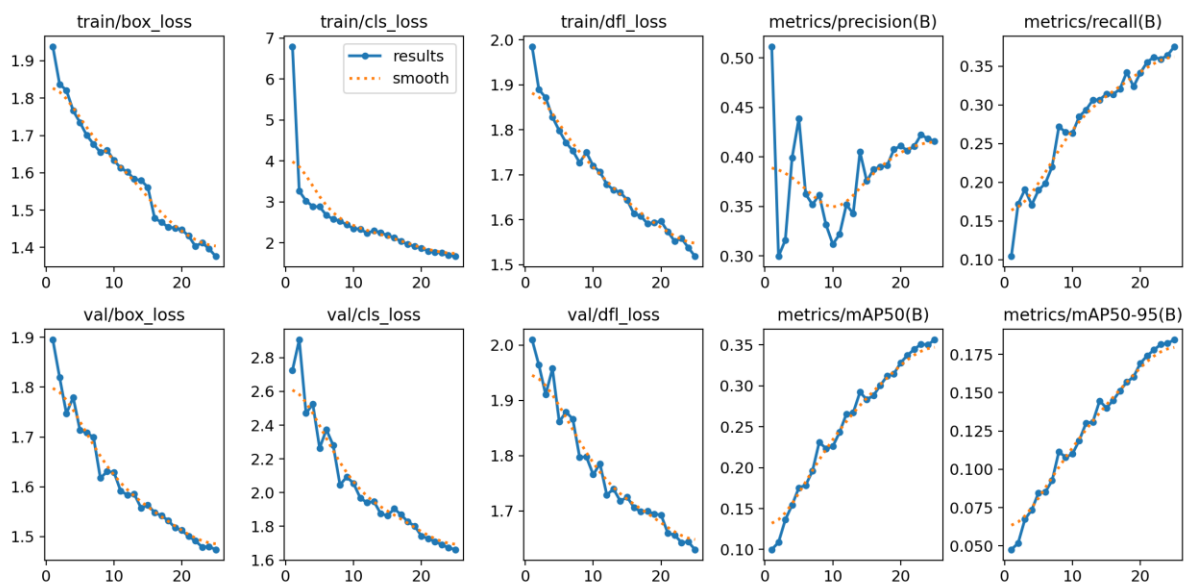
Input Size: 1204*1204

Echos:25

Batch Size: 8

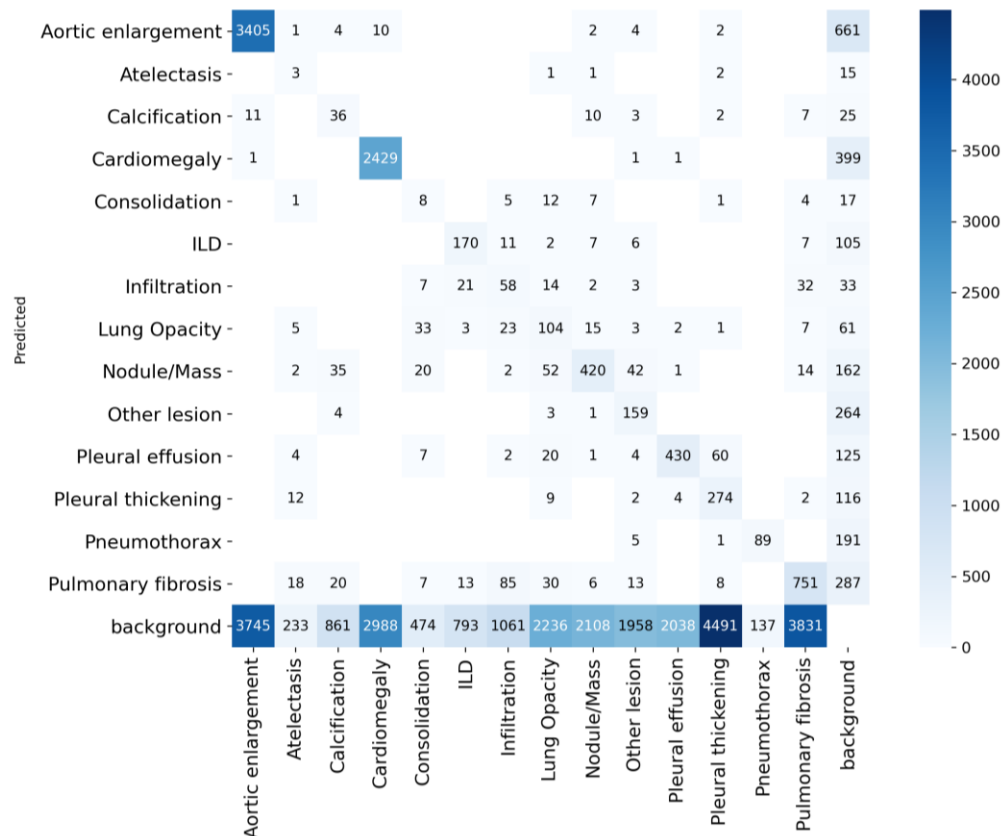
Classes: 14 disease categories

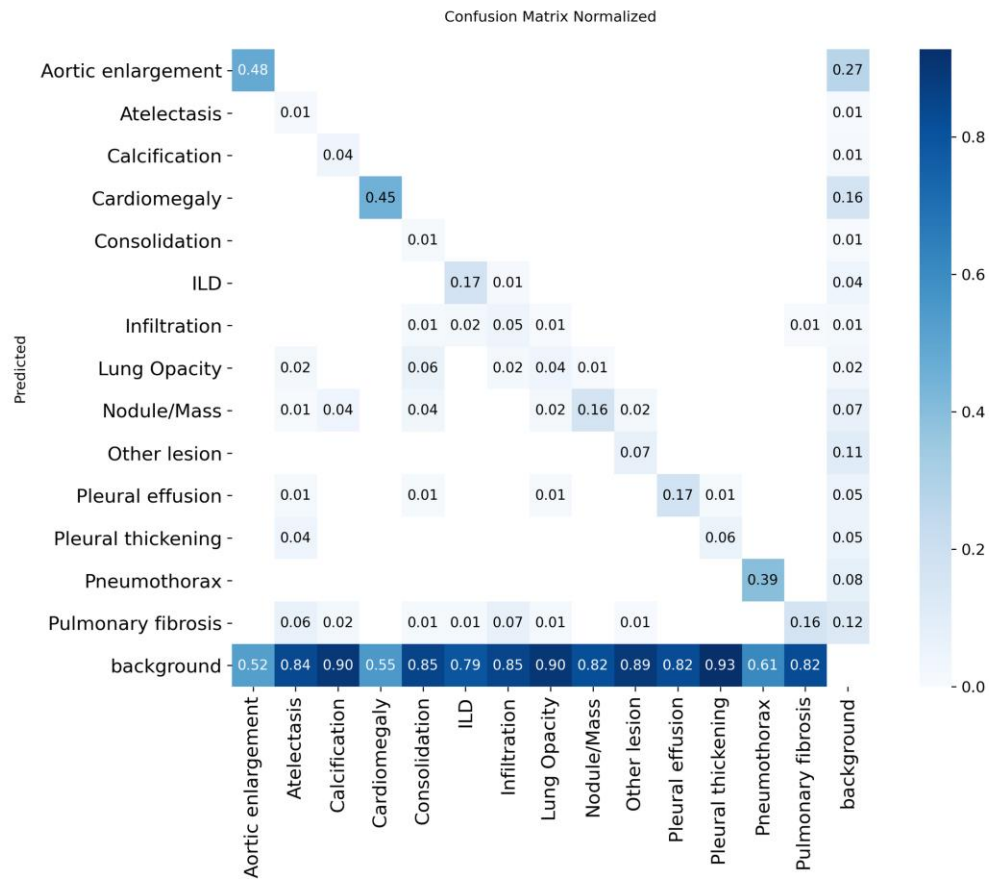
Training Visualizations:



5. Classification Model (ResNet50)

For the classification task, a ResNet50 model pretrained on ImageNet was implemented. Approach: To establish a baseline, the multi-label dataset was converted into a single label classification task by selecting one primary disease per image





6. System Workflow & Model Saving

The integrated pipeline for CliniScan is structured as follows:

Input: Chest X-ray Image

Detection: YOLO model locates the abnormal region

Classification: ResNet model predicts the specific disease type

Saved Artifacts:

- Detection Model: cliniscan_detection_model.pt
- Classification Model: cliniscan_classification_model.pth

Challenges Faced:

- Handling multi-label data and class imbalances.
- GPU limitations within the colab environment.
- Incorrect bounding box normalization

7. REFERENCES

- Vin-CXR Dataset
- Ultralytics YOLO Documentation
- Py Torch Documentation